

CTMT as a Single Machine

Synthesis, the Life-Cycle of Coupling, and a One-Knob Demonstration that the Layers
React to One Another

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Abstract

This capstone does not add a theory; it shows that the CTMT layers are one machine. All of them—resolved/null decomposition, covariance blocks, admissible automorphisms, Fisher/response geometry, information content, estimation efficiency, classification, and dynamical transport—are derived structures of a single primitive $\mathcal{G} = (O, g; R, \Sigma)$ and its linear-stochastic extension. We identify the through-line that binds them: the resolved–null coupling C_{RN} . Its *life-cycle* is the rigorous meaning of “seepage”: it is *born* by the cross-drift (generation equation), *propagates* as a Mori–Zwanzig memory kernel, is *measured* by the sector mutual information, *pays off* as an estimator efficiency gain, *constrains* the symmetry group by rigidity, *classifies* the geometry as an irreducible coordinate, and *globalizes* as holonomy. Crucially these are locked by exact identities—most sharply, the estimation efficiency gain equals the information content, $\frac{1}{2} \log(\det V_{\text{resolved}} / \det V_{\text{full}}) = I(y_R; y_N)$ —so the layers are not analogies but the same quantity seen from different sides. We demonstrate this with a single controlled system: turning one knob (the cross-drift strength) makes coupling appear and drives every layer’s diagnostic in lockstep, with the identity holding throughout (verified numerically). Finally we retire, with reasons, the unsupported physical reading of the earlier Fisher-rank-loss / Navier–Stokes draft, and state the defensible residue. The machine is tight; its physical interpretation remains the open frontier already mapped elsewhere in the programme.

Contents

1 Purpose: from a wall-tap delay to a machine	1
2 One primitive, many derived layers	2
3 The through-line: the life-cycle of coupling	2
4 One-knob demonstration: watching the machine react	3
5 What is retired, and the defensible residue	4
6 Conclusion	5

1 Purpose: from a wall-tap delay to a machine

The programme began with a physical intuition—an unresolved influence reaching an observable with a delay, a “wall-tap” echo—and passed through many named geometries (collapse,

transport, modulation, atlas, null leakage). The honest lesson of the hardening was twofold. Where the layers made *mathematical* claims, they could be made rigorous and, it turns out, they fuse into one object. Where they reached for *physics*—turbulence, Kolmogorov scaling, nodes of presence via Fisher-rank loss—the claims outran the mathematics and must be retired (Section 5).

This paper serves the first half. Its thesis is structural, not physical:

CTMT is not a bundle of separate constructions. It is a single machine built on one primitive, whose layers share and react through one through-line, the resolved-null coupling, and are locked together by exact identities.

Non-claim 1.1 (No new theory, no physics). Nothing here is a new theorem beyond assembling and cross-linking the proven results of the foundations, covariance, atlas/null-sector, dynamics, and pillar papers, plus one demonstration. No physical claim is made or revived.

2 One primitive, many derived layers

Definition 2.1 (The primitive and its dynamical extension). The static primitive is the resolution geometry $\mathcal{G} = (O, g; R, \Sigma)$: an observation space with reference metric, a resolved sector $R = \text{Im}(J)$, its g -orthogonal null sector $N = R^\perp$, and a covariance Σ with blocks $(\Sigma_R, \Sigma_N, C_{RN})$. The dynamical extension adds a generator A via the linear-stochastic law $dy = Ay dt + B dW$, $Q = BB^\top$.

Every layer of the programme is a function of these data; none is introduced independently.

Layer	Derived object	Source
Decomposition	$O = R \oplus N$, projectors	Foundations
Covariance geometry	$\Sigma_R, \Sigma_N, C_{RN}$	Covariance
Response/Fisher	$J^\top \Sigma^{-1} J = J_R^\top S_R^{-1} J_R$	Covariance
Information	$I(y_R; y_N)$ via canonical correlations	Atlas/Null
Estimation	coupling-aware GLS efficiency	Pillar 1
Symmetry	$\text{Aut}(\mathcal{G})$ (coupling stabilizer)	Covariance/Atlas
Classification	moduli $rq+r+q$, coupling irreducible	Pillar 3
Dynamics	Lyapunov flow, memory kernel	Dynamics
Global	atlas, holonomy, curvature	Atlas/Dynamics

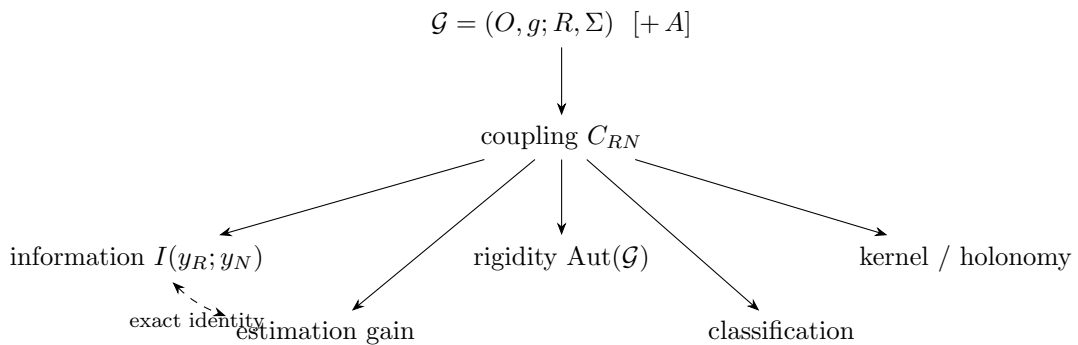


Figure 1: The dependency spine. Every layer is a function of the one primitive; the coupling C_{RN} is the shared through-line, and the information and estimation layers are tied by an exact identity (Theorem 3.1).

3 The through-line: the life-cycle of coupling

“Seepage” has been used loosely across the drafts. We fix its rigorous meaning here as the *life-cycle of the coupling* C_{RN} , and trace it through the layers, each step a proven result of the programme.

1. **Born (dynamics).** From a decoupled state the coupling is generated at first order by the cross-drift, $C_{RN}|_{C_{RN}=0} = A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top$.
2. **Propagates (dynamics).** Eliminating the null sector yields the exact memory kernel $K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}$: the delayed feedback of the unresolved sector into the resolved one. This is the matured form of the original wall-tap intuition.
3. **Measured (information).** The coupling carries information $I(y_R; y_N) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2)$ in the canonical correlations ρ_i of the whitened coupling.
4. **Pays off (estimation).** A coupling-aware estimator is strictly more efficient than a resolved-only one, by the Schur-complement amount, when the null baseline is calibrated (Pillar 1).
5. **Constrains (symmetry).** Nonzero coupling imposes the rigidity constraint $AC_{RN}B^\top = C_{RN}$, shrinking $\text{Aut}(\mathcal{G})$ from a product group to a coupling stabilizer.
6. **Classifies (invariants).** Coupling is an irreducible coordinate of the moduli space, occupying rq of its $rq + r + q$ dimensions (Pillar 3); it cannot be recovered from the block spectra alone.
7. **Globalizes (atlas).** Across charts, coupling transport around loops yields holonomy in $\text{Aut}(\mathcal{G})$; nontrivial holonomy classifies nontrivial global geometries.

That these are one machine, not seven analogies, is shown by the exact links between them. The sharpest is between information and estimation.

Theorem 3.1 (Information equals estimation gain). *In the just-identified linear-Gaussian chart with calibrated null baseline, the log-volume reduction of the Cramér–Rao confidence ellipsoid equals the sector mutual information:*

$$\frac{1}{2} \log \frac{\det V_{\text{resolved}}}{\det V_{\text{full}}} = -\frac{1}{2} \sum_i \log(1 - \rho_i^2) = I(y_R; y_N).$$

Proof. With $J_R = I$, $V_{\text{resolved}} = \Sigma_R$ and $V_{\text{full}} = S_R = \Sigma_R^{1/2}(I - \tilde{C}\tilde{C}^\top)\Sigma_R^{1/2}$, so $\det V_{\text{full}}/\det V_{\text{resolved}} = \det(I - \tilde{C}\tilde{C}^\top) = \prod_i (1 - \rho_i^2)$; compare the Gaussian mutual-information identity. (Pillar 1, Corollary; verified numerically across the demonstration below.) $\square$

A second link ties the dynamical and static faces: the same cross-drift blocks (A_{RN}, A_{NR}) that generate the static coupling (step 1) also constitute the memory kernel (step 2), so a single dynamical cause drives both the transient and the stationary machine.

4 One-knob demonstration: watching the machine react

We now let a reader *see* seepage. Take one Ornstein–Uhlenbeck system $dy = Ay dt + B dW$ on $O = \mathbb{R}^2 \oplus \mathbb{R}^2$ with fixed within-sector drift and isotropic noise, and a single knob: the strength α of the cross-drift blocks $A_{RN} = \alpha A_{RN}^0$, $A_{NR} = \alpha A_{NR}^0$. At each α we solve the algebraic Lyapunov equation for the stationary covariance and read off every layer’s diagnostic. Because

the same α sets both the memory kernel and, through the Lyapunov solution, the stationary coupling, one knob drives the entire machine.

α	$\ C_{RN}\ _F$	$I(y_R; y_N)$	Fisher gain	vol. ratio	kernel norm	dim Aut	identity
0.00	0.000	0.000	0.000	1.000	0.000	2	✓
0.15	0.036	0.006	0.042	0.988	0.014	0	✓
0.30	0.073	0.025	0.168	0.952	0.056	0	✓
0.50	0.128	0.070	0.461	0.870	0.155	0	✓
0.80	0.233	0.188	1.143	0.687	0.396	0	✓
1.10	0.406	0.387	2.070	0.461	0.749	0	✓

Table 1: One knob α ; the whole machine reacts. Coupling $\|C_{RN}\|$ is born from zero; mutual information, Fisher gain, and dynamical memory-kernel norm rise together; the estimator confidence-volume ratio falls (tighter estimates); the automorphism dimension collapses from 2 to 0 the instant coupling appears (rigidity); and the information=estimation identity of Theorem 3.1 holds at every α . Values from a verified computation.

Remark 4.1 (What the table shows). Three things are visible at once. *Sharing*: no diagnostic moves alone; a change in the single cross-drift propagates to every layer. *Reacting*: the symmetry layer responds discontinuously—any nonzero coupling kills all continuous automorphisms (dimension $2 \rightarrow 0$), the exact rigidity predicted by the stabilizer constraint. *Locking*: the information and estimation layers are not merely correlated but equal, at every knob setting. This is the concrete face of “a tight single machine.”

5 What is retired, and the defensible residue

Honesty about the machine requires retiring the parts that were physics-by-name. The earlier Fisher-rank-loss draft (“seepage as Fisher rank loss; nodes of presence; Navier–Stokes; Kolmogorov”) cannot stand as written, for concrete reasons.

1. **Two different “seepages.”** That draft defines seepage as Fisher *rank loss* $\lambda_{\min}(F) \rightarrow 0$. That is a distinct phenomenon from the coupling-seepage of this programme ($C_{RN} \neq 0$); conflating them is a terminology hazard. Rank loss is loss of identifiability; coupling is cross-sector correlation. Only the latter has the life-cycle of Section 3.
2. **The turbulence theorems are sketches.** “Necessity of constraint emergence” and “localization under global rank loss” rest on an unproven “conservation of observable variance/sensitivity”; no such conservation law is established, so the conclusions (spectral concentration, forced localization) do not follow.
3. **The Kolmogorov recovery is circular and numerically inconsistent.** The synthetic construction *assumes* $\lambda_k \propto k^{-4/3}$ and reports $E(k) = k\lambda_k \propto k^{-5/3}$; but $k \cdot k^{-4/3} = k^{-1/3}$, not $k^{-5/3}$, so the exponent bookkeeping does not close. More fundamentally, assuming the scaling and then reading it back is circular—the same representability trap identified for the Omori attempt (Pillar 2): a flexible spectrum can host any power law, so matching one confirms nothing.
4. **Nodes of presence are postulated, not derived.** The three joint conditions (curvature maximal, Fisher bounded below, hazard minimal) are asserted; the existence theorems are hand-waved.

Remark 5.1 (Salvageable, deflated residue). Two honest kernels survive. *Fisher conditioning is a legitimate diagnostic*: ill-conditioning of $J^\top \Sigma^{-1} J$ genuinely signals loss of local identifiability, and tracking $\lambda_{\min}$ and κ is standard and useful—it simply does not derive turbulence. And a *node of presence has a rigorous deflated meaning*: the persistently well-conditioned resolved subspace, i.e. the directions where the Fisher eigenvalues stay bounded below as others degrade. That is exactly R (or its well-conditioned part) and is already in the machine. Kept at this strength, both are defensible; pushed to physics, neither is.

6 Conclusion

The journey from a wall-tap delay to a morphism programme produced, on the mathematical side, not a scattering of geometries but one machine. A single primitive $\mathcal{G} = (O, g; R, \Sigma)$, with a linear-stochastic generator, yields every layer as a derived structure; the resolved–null coupling C_{RN} is the shared through-line; and its life-cycle—born by the cross-drift, propagated as a memory kernel, measured as information, cashed as estimation efficiency, enforced as rigidity, irreducible in classification, and globalized as holonomy—is the rigorous content of “seepage.” The layers are locked by exact identities, the sharpest being that the estimation gain *equals* the information content; and a single knob drives the entire apparatus in lockstep, as demonstrated. What does not belong to the machine—the Fisher-rank-loss turbulence reading—has been retired with reasons, leaving a defensible residue.

The construct is therefore what it was hoped to be: a tight, single, layered machine whose parts share and react. Its ultimate physical status is not settled here and is not claimed; that frontier has a precise address elsewhere in the programme (the dynamical buildup ladder and the fluctuation–dissipation cross-exponent test). What can be said now, and defended, is that the machine is one machine.

References

- [1] M. Rada, *Resolution Geometry of Observable Transport Systems* (CTMT foundations), 2026.
- [2] M. Rada, *CTMT Covariance Geometry*, 2026.
- [3] M. Rada, *CTMT Atlas and Null-Sector Information*, 2026.
- [4] M. Rada, *CTMT Dynamics: Linear-Stochastic Layer and the Mori–Zwanzig Seepage Kernel*, 2026.
- [5] M. Rada, *Coupling-Aware Estimation in CTMT: a Cramér–Rao Efficiency Gap*, 2026.
- [6] M. Rada, *Complete Invariants of CTMT Covariance Resolution Geometry*, 2026.
- [7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- [8] R. Zwanzig, “Memory effects in irreversible thermodynamics,” *Phys. Rev.* 124 (1961), 983–992.